

“Empowering Science and Data Analytics through Python”

Assignment On

Lecture_02

Topic:

Conditionals and Loops:

- i) If**
- ii) elif**
- iii) else,**
- iv) For**
- v) while,**
- vi) break**
- vii) continue**

1. if , elif, else

1. Write a program to check if a number is positive.
2. Check if a number is even.
3. Find the largest of three numbers.
4. Write a code that take a user input: **marks**. If marks is greater than 75 then print output "Distinction", if marks is greater than 40 then print output "Pass". If less than 40 then print output "Fail".
5. Shyam is of age 25. He wants to play a computer game. The game has some initial inquiry asking the user to input the age. Use condition that
 - i) if age is greater than equal to 20 print output will be "Shyam can play".
 - ii) if age is less than 20 print output will be "Shyan cannot play".
 - iii) if age is something else then "Enter a valid number".
6. Take a use input an integer and print output "Greater" if the number is greater than 5. If the number is any other number including 5 then "smaller".

2. for and while

Application of loop:

Practice:

For loop:

1. Print numbers from 1 to 10.
2. Print even numbers from 1 to 20.
3. Print sum of first 10 numbers.
4. Count how many numbers are divisible by 3 from 1 to 30.

while loop:

1. Print numbers from 1 to 10.
2. Print even numbers from 1 to 20.
3. Print multiplication table of 5.
4. Count digit in a number.

break and continue

Practice:

break

1. Stop printing when number = 5.
2. Find the first number divisible by 7.
3. Stop when negative number is entered.
4. Find first even numbe in a list.

Continue

Practice:

1. Skip number 5 while printing 1 to 10.
2. Print only odd numbers.
3. Skip negative numbers in a list.
4. Skip multiples of 3.

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